

Reinventing Institutions: Bricolage and the Social Embeddedness of Natural Resource Management

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I. INTRODUCTION

This study questions the idea that appropriate mechanisms can be designed to ensure optimum resource use, beneficial collective action and hence to build social capital. I argue here that the school of 'institutional crafting' in natural resource management is based on concepts which are inadequately socially informed and which ill-reflect the complexity, diversity and *ad hoc* nature of institutional formation.

Three aspects of institutional bricolage are illustrated here: the multiple identities of the bricoleurs; the frequency of cross-cultural borrowing and of multi-purpose institutions; and the prevalence of arrangements and norms which foster co-operation, respect and non-direct reciprocity over lifecourses.

In elaborating the concept of bricolage, I raise questions about whether local institutions are amenable to design, the scope for negotiating the norms which underlie institutional arrangements and the extent to which different institutions may be emancipatory or exclusionary. I conclude that development interventions aimed at institution building should be based on a socially informed analysis of the content and effects of institutional arrangements, rather than on their form alone.

II. CONFLICTS OVER RESOURCES IN THE USANGU BASIN

Planners and policy-makers perceive the Usangu Basin as facing problems of degradation and depletion of its grazing land and water resources, due to multi-user competition, and an institutional deficit resulting in open access resource use. Competition over natural resources is a critical concern because the severe drying up of the Ruaha river has potentially deleterious effects on the perennial swamp (the *ihetu* – an area of precious biodiversity), on hydro-electric power

generation, on irrigated agriculture and on the wildlife tourism potential of the downstream Ruaha National Park. A DfID funded project (Sustainable Management of the Usangu Wetland and its Catchment – or SMUWC) has been investigating the causes of resource depletion in Usangu and developing local capacity to collectively manage the natural environment.

The Usangu basin is characterised by a long history of in-migration (of Baluchis from Iran, Masaai and Sukuma pastoralists from the north of Tanzania and agriculturalists from neighbouring areas). Whilst potentially competing groups of users in the Usangu basin are various, the most common characterisation of this competition is as one between ethnically based groups of sedentary agriculturalists (the 'indigenous' Sangu and in-migrants such as Nyakusa and Hehe) and itinerant pastoralists (predominantly Il Parakuyu Masaai and Sukuma). Notably, 'ethnic' agriculturalists predominate in local political and administrative structures. Often implied in policy discussions is a developmental struggle between entrepreneurial 'modernising' agriculturalists and intransigently 'backward' pastoralists.

In project and policy documents and in the discourse of development at district and regional level natural resources are frequently characterised as 'fragile' and 'depleted' by conflict ridden over-use, the cause of degradation being their open access status caused by the erosion of traditional authorities and ever-increasing population pressures [SMUWC, 2001a].

Local bureaucratic village institutions are characterised at best as ineffectual (through poor communication, high turnover of officers and lack of resources), at worst as corrupt and rent seeking. Village councils are seen as badly organised, informed and equipped, lacking transparency and often dominated by a few powerful people [SMUWC, 1999, 2001a]. Indigenous or traditional resource management arrangements are largely unrecognised or thought to be non-existent. There is then a perceived management deficit in terms of the organisation of collective action and the control of natural resources.

Crafting Institutional Solutions

Despite the 'conflict and corruption' analysis of existing institutions, policy and project documents assert the 'natural' basis of co-operation latent in village life and put great faith in the efficacy of *new* formal institutional arrangements to support collective resource management [SMUWC, 2001b]. The Village Land and Natural Resource Management Committees proposed in original project documents (subsequently implemented as Environment Committees) epitomise ideas commonly expressed in institutional theory. They are intended to operate 'in a formal and transparent way if they are to work effectively and to win public confidence'. Their purpose is to assess land and water use trends, introduce land registration, formulate village land use plans, allocate land and water rights, draft by-laws, identify and resolve

conflicts. The committees are intended to be representative, consisting of 'a reasonably small group' to facilitate consensus on complex issues and to make it easier to provide training. The village committees are to be linked with other layers of resource management arrangements, in particular through interaction with district level natural resource management teams [SMUWC 1999].

Such prescriptions clearly derive from theories about the type of 'robust' institution suited to resolving common property resource management dilemmas. A focus on formal public structures with clear boundaries, transparency, representativeness and the codification of rules through written bye-laws, contracts and the specification of property rights is common to the literature on 'design principles' for institutional development. Ideas about the benefits of small and relatively homogeneous groups of decision-makers (usually representing 'a community') neatly linked or 'nested' within layers of structures (for example, at district national or regional level) and the possibility of such groups identifying and implementing an optimum level of resource use are also prevalent [Bromley and Cernea, 1989; Ostrom, 1990, 1992, Wade, 1988, Agarwal, 1997].

How likely are such new institutions, established with a strong focus on formalisation, transparency, representation, regulation and rights, to provide 'the answer' to collective resource management in Usangu?

III. INTERROGATING INSTITUTIONAL THEORY

There are a number of grounds on which the faith placed in such prospective arrangements can be criticised [Cleaver, 2000]. These include simplified and unilinear models of institutional evolution, the assumed primacy of narrow productive concerns amongst resource users and a purely instrumental view of culture and social structure as a 'resource bank' from which social capital is drawn to facilitate the implementation of good resource management decisions.

A short digression on the terminology of institutions is appropriate here. In rejecting dichotomous classifications of institutions as either formal or informal, traditional or modern, it is difficult to find alternative labels without reproducing false polarisations. In this analysis I generally use the terms 'bureaucratic' and 'socially embedded' to distinguish between institutions. Bureaucratic institutions are those formalised arrangements based on explicit organisational structures, contracts and legal rights, often introduced by governments or development agencies. Socially embedded institutions are those based on culture, social organisation and daily practice, commonly but erroneously referred to as 'informal'. It should be clear from my argument in this study that the two are not necessarily easily distinguishable; 'bureaucratic' institutions may be 'socially embedded', but are not inevitably so, while

processes of bricolage may result in the bureaucratisation of 'traditional' cultural or social arrangements.

Institutional theory and much policy tend to favour an analysis of institutions as either 'weak' or 'robust' and to attribute normative value to bureaucratic arrangements, partly because of their visibility and legibility to development agencies. Weak institutions can be transformed into strong ones by resource users and policy-makers through an active process of institutional design and crafting [Ostrom, 1992: 60]. Ostrom sees crafting as a continuous evolutionary process of developing the optimal institution for the job in hand. Culture and social structure then becomes another raw material, part of the institutional resource bank from which arrangements (especially distributional norms and relations of trust) can be drawn which reduce the 'social overhead costs' of co-operation. Increasingly such resources are referred to as 'social capital' but as Ostrom herself admits there is generally a lack of understanding about how to 'create, maintain and use social capital' [Ostrom, 1992: 23].

Ideas about the strategic use of social capital are not confined to institutional theory, but have become widespread in development thinking. For example, in common with institutional theory, sustainable rural livelihoods (SRL) literature conceptualises social relations as a potential store of assets upon which people can draw to construct effective livelihood strategies. SRL literature broadly defines social capital as resources of reciprocity and trust, which can be drawn upon by households in the composition of sustainable livelihoods. Social relations, institutions and organisations are seen as critical mediating mechanisms as they enable and constrain the actions of individuals and households [Scoones, 1998; Ellis, 2000]. Substantial stocks of social capital are seen as necessary in generating wealth both for households and communities, in ensuring beneficial collective action and common property resource management and the proper working of local administrative and political structures [Narayan, 1997; Uphoff and Wijayarathna, 2000].

Such instrumentalist views assume individual actors are political and social entrepreneurs who rationally utilise social capital to craft institutions in pursuit of optimal resource management. Interactions over resource management then become modelled as transactions about which the individuals make the appropriate calculations of cost and benefit, based on single preferences. Because of the emphasis on tangible and identifiable behaviours and incentives, there is a tendency towards attributing greater value to formalised modes of interaction and codified norms.

However, it is often claimed that in order to properly utilise social capital, institutions must be appropriately 'embedded' in the social milieu from which the norms to support purposive decision making can be drawn [Ostrom, 1990]. Such concepts of embeddedness imply a state of solidity and fixedness; the firm location of institutions in the social environment. Rather than seeing

dynamic social relations and the changing cultural milieu as the very stuff of people's lives, they are seen as a social cement which can be consciously utilised to strengthen institutions. These rather mechanical views of social embeddedness clearly link with the formalised arrangements preferred by institutionalists. However, they throw little light on processes of institutional evolution, which I suggest are more ad hoc, approximate and shaped by social life and culture than is implied by concepts of design and crafting. In much institutional theory, social capital is viewed as an undoubted good and institutional change as occurring along unilinear trajectories of evolution towards a robust state, or deterioration, dysfunction and collapse. These assumptions, reinforced by the terminology of the market (investment, capital, costs, transactions), obscure rather than elucidate the relational nature of collective action, the ebb and flow in social life of relations of trust and association across lifecourses, and the mix between conscious and unconscious motivation and action on the part of agents.

In this study I outline an alternative approach to conceptualising institutions through understanding processes of bricolage. I show how a variety of socially embedded institutions for collective action exist in Usangu in addition to the newly introduced bureaucratic resource management structures. Institutions of co-operation are embedded in everyday relations, networks of reciprocity and the negotiation of cultural norms. Through processes of bricolage these may combine with or replace contracts, legal rights and formal sanctions; without such bricolage and the social embedding of new arrangements, bureaucratic institutions are unlikely to be effective.

IV. INSTITUTIONAL BRICOLAGE

An alternative approach to institutions is needed, which allows us to look beyond the formal organisations and to conceptualise social relations as more central than simply context or assets. Drawing on the work of Douglas [1973, 1987] Peters [1994] and Giddens [1984] as well as my own previous work on resource management in Zimbabwe [Cleaver, 2000], I suggest that the concept of institutional bricolage enables us to do this.

Douglas elaborates Levi-Strauss's concept of 'intellectual bricolage' [Douglas, 1987: 66] and extends it to institutional thinking to illustrate how the construction of institutions and decisions to act are rarely made on the basis of individual rational choice. Instead 'institutions do the thinking' on behalf of people and institutions are constructed through a process of bricolage – gathering and applying analogies and styles of thought already part of existing institutions. Symbolic formulae are used repeatedly in the construction of institutions, thereby economising on cognitive energy by offering easy classification and legitimacy [1984: 76]. She emphasises the sameness and

constraint of this form of institutional development; 'The bricoleur uses everything there is to make transformations within a stock repertoire of furnishings' [1987: 66]. In earlier work Douglas considers the concept of institutional leakage: 'Sets of rules are metaphorically connected with one another, allow meaning to leak from one context to another along the formal similarities that they show' [Douglas, 1973: 13]. This suggests a less conscious and less rational/functional formation of institutions than that proposed by many authors writing of the institutions of common property resource management.

Adapting Douglas and Levi-Strauss I use the term 'institutional bricolage' to suggest how mechanisms for resource management and collective action are borrowed or constructed from existing institutions, styles of thinking and sanctioned social relationships. This raises the question of the critical role of human agency in shaping and reshaping institutions; how far do individuals act as institutional engineers? Douglas suggests that the constraining aspects of socially embedded institutions seriously circumscribe the scope for individual thought and action: 'Institutions systematically direct individual memory and channel our perceptions into forms compatible with the relations they authorise' [Douglas, 1987: 92]. A less constrained view of the individual's role in institutional formation is put by Giddens [1984] who sees the individual as a possessor of agency as well as specific and often multiple social identities which may change over lifecourses. Different *bricoleurs* are thus likely to apply their knowledge, power and agency in respect to social relations, collective action and resource management in differing ways. The result is a rich diversity of pliable, if approximate, institutional arrangements.

Building on both Douglas and Giddens, I reject a view of individuals as rational and essentially economic resource appropriators, and attempt to reconceptualise them as conscious and unconscious social agents, deeply embedded in their cultural milieu but nonetheless capable of analysing and acting upon the circumstances that confront them. Individual action is characterised both by agency and structural constraint. For example I have previously illustrated in an examination of water management in Zimbabwe [Cleaver, 2000] how people consciously reduce the cognitive effort of responding to change by drawing on and adapting existing norms and mechanisms to new purposes. Less consciously, the use and adaptation of pre-existing customs and practices confer new arrangements with the legitimacy of 'tradition'; they become part of the 'right way of doing things'. Many institutional arrangements are forged in practice through daily interactions, the necessary improvisation involved in social life [Bourdieu, 1977] and the constant use of resources.

I suggest that it is a false dichotomy to pose a realm of 'traditional' informal, culturally and socially embedded institutions against a 'modern'

domain of rationally designed committees and formal structures, and to suggest that one is likely to be better than the other at resolving conflicts or managing natural resource use. Local resource management arrangements are a complex blend of formal and informal, traditional and modern. The evolution of collective decision-making institutions may not be the process of conscious selection of mechanisms fit for the collective action task (as in Ostrom's model) but rather a messier process of piecing together shaped by individuals acting within the bounds of circumstantial constraint. Institutions so derived may survive partly due to the legitimacy bestowed by 'tradition', the moral command of what went before over the present [Giddens, 1984]. Institutions are formed through processes of bricolage in which similar arrangements are adapted for multiple purposes, are embedded in networks of social relations, norms and practices and in which maintaining social consensus and solidarity may be equally as important as optimum resource management outcomes.

The concept of bricolage helps us to examine the complex and dynamic nature of natural resource management and the 'fit' between institutions and the web of livelihood networks and practices in which they are embedded. I will proceed to elaborate processes of institutional bricolage by illustrating three aspects of bricolage with selected data from Tanzania; the complex identities and norms of the bricoleurs, the practice of cultural borrowing and adaptation of institutions to multiple purposes and the prevalence of common social principles which foster co-operation (as well as conflict) between different groups of stakeholders.

V. AGENTS, IDENTITIES AND THE ARTICULATION OF NORMS

Rejecting the idea of narrowly rational 'institutional engineers' in favour of the socially embedded 'do-it yourself' bricoleurs, we then need to socially locate such agents. Indeed Granovetter suggests that the social and historical embeddedness of institutions manifests through networks of personal relations [Granovetter, 1992: 62]. Institutional theory is often deficient in investing resource users with any meaningful social identity. It generally emphasises productive identities (such as 'irrigators', 'pastoralists') and a very limited number of social roles ('leaders', 'women'). People participating or represented in formal institutions are commonly assumed to have overriding productive incentives and/or clear social roles which render them fit for the task. In this model there is a key role for community representatives in shaping the institutions, mediating the social and cultural norms, producing and interpreting rules and enforcing sanctions. The emphasis on such participators is problematic as formal institutions often reproduce existing patterns of inequity (in Usangu inequalities of wealth, ethnicity and gender) and may serve to shape and reinforce other differences.

Livelihood Flexibility and Lifecourses

The privileging of single aspects of people's identities for institutional purposes is problematic as it ill reflects complex social and livelihood identities and multiple motivations. Nor does it reflect the importance of temporality in the formation of identity-people's relationships, their productive and reproductive concerns change dramatically during their lifecourse with neither economic activities nor ethnic identities alone adequately reflecting their interests. For example, in Usangu people's interests rarely fit easily into the agriculturalist/pastoralist divide; many pastoralists are semi-sedentarised and engaged in cultivation, whilst young pastoralist men migrate or mine gold in order to establish themselves economically. Similarly, as agriculturalists generate surplus they may invest in cattle, and use these to marry, support extended families, diversify into business. If people's identities are dynamic then the social materials and norms which shape institutions are likely to be equally difficult to categorise. An example of overlapping identities generating complex norms and networks is provided by one interviewee, Karim, a farmer of rainfed maize and groundnuts. He also engages in business; buying and selling charcoal and buying rice when the price is low, storing and selling when it is high. He is chairman of his hamlet and chairman of the water committee. He is a Sangu by ethnic origin and a traditional healer (herbalist). He follows Sangu traditions and ceremonies, particularly emphasising the worship of ancestors, in order to ensure the well-being of the family and productivity of natural resources. He is also a Muslim and he and Islamic leaders gather at his home for prayers and the appropriate Islamic traditions and festivals.

In emphasising particular identities and roles, formal institutionalism may not just reproduce but reinforce or amplify social divisions. This is particularly so when a static interpretation of 'culture' is adopted and incorporated into bureaucratic institutions as illustrated by Maganga who shows how the codification of customary law in Primary Magistrates courts and Ward Tribunals, may emphasise ethnic differences in Usangu [Maganga, 1999; Maganga and Juma, 1999].

Representation, Articulation and Exclusion

Institutional theory and policy implies that representatives on committees and in associations directly represent the resource users by translating community norms into more regularised institutional arrangements [SMUWC, 2001a, 2001b]. However, norms and practices, and the relationships of trust and co-operation which underlie them, are often generated and negotiated outside the formal institutions. Institutional bricolage then, takes place in a wider arena than that defined by the visible structures of bureaucratic resource management institutions. Evidence from Usangu illustrates the diverse

location of decision-making, the importance of households and wider social networks in the generation of norms and practices of resource use. For example, children and young people play a major role in resource use and management through practice in Usangu, although they have no place in bureaucratic institutions. Children as young as three herd small livestock whilst older children and 'youth' make decisions about where water or feed herds and are explicitly consulted by their families over matters of livestock welfare, grazing and the implications for cattle condition. In the dry season children and young people may stay away from home in the grazing camps at the *ihefu* for months at a time, managing their animals and their use of pasture and water, and participating in a form of collective cattle protection militia. The key role of children in resource use and their complete lack of inclusion in bureaucratic management structures raise questions about the mediating processes between the creation of rules-in-use and their development into collective arrangements. The concept of institutional bricolage more adequately allows us to reflect the diverse location of the generation of institutional arrangements.

Bricolage is an authoritative process and some 'bricoleurs' are likely to possess more authoritative resources than others. Overlapping social identities mean that people may call on a variety of attributes to justify institutional position or influence. These vary from economic wealth, specialist knowledge or official position, to kinship and marriage, or personal characteristics such as eloquence, strength and honesty – a point well illustrated by Dikito-Wachtmeister [2000] in relation to women's participation in water resource management in Zimbabwe. Multiple sources of authority mean inevitable negotiation about who has the right to form the rules.

An example of the complexity of authority, articulation and participation in processes of bricolage is found in the role of women in resource management in Usangu. Women interviewed often stated that they felt able to participate fully in meetings to discuss resource management at hamlet level, but only to observe at wider village meetings, where they felt they 'did not have the language' to contribute. None the less women were significant managers of resources in practice, even in areas presented to us as 'traditionally' the concern of men, like cattle production. Women in fact played major roles in cattle breeding, in caring for sick cattle, preparing medicines, breeding and in decisions about milk production and cattle sales [King and Ole-Lengisugi, 1999]. Evidence from household interviews suggests that in the majority of cases, both men and women regarded decisions over natural resource use as properly made by all adult members of the family. However, both traditional cultural mechanisms for managing grazing and formal bureaucratic organisations of cattle keepers exclusively involved adult men. How far the decisions made in households and the resulting practices coincide with or

contradict the rules and norms being produced in more formalised institutional settings is a moot point and illustrates the diverse location of processes of bricolage.

Bureaucratic institutions created through design and socially embedded arrangements formed through bricolage may both reinforce and perpetuate social divisions. Despite the possibility that new bureaucratic institutions could be emancipatory in including people previously excluded, in Usangu the poorest people rarely participate in such fora (even if formally eligible to do so) due to extreme time and labour constraints. Moreover, the poorest families are characterised by social isolation and very weak networks, so impeding their ability to contribute to processes of bricolage through daily social interaction. The poorest families in our study received little or no support from their neighbours who characterised them as 'unlucky' or 'lazy'. If they had been unfortunate enough to have suffered a number of family bereavements, particularly of children, surviving members were often suspected of witchcraft and treated with suspicion [SMUWC, 2001c].

A recognition of differences in social power amongst resource users necessitates a focus on the daily interactions and beliefs upon which such inequities are based, if we are to avoid reproducing these in new bureaucratic institutions.

Norms and the Limits to Negotiation

The concept of bricolage implies an active assembly of parts, the adaptation of norms, values and arrangements to suit a new purpose. It implies both a conscious scrutiny of some beliefs, and an unconscious acceptance of others in the formation of institutions. The simultaneous acceptance and questioning of traditions can be illustrated by the case of Rahel, a Maasai woman who is also a born again Christian (Pentecostalist) and an elder of the church. She belongs to a Maasai women's choir which functions as a women's support group and all members of the household take part in a collective labour group of ethnically mixed neighbours for agricultural work. Her husband has not converted to Christianity and is the hamlet leader and a leading member of local and national Maasai cultural and political structures. Rahel sees both advantages and disadvantages to the household of their diverse cultural and social networks. Her strong Christian beliefs lead her to question certain manifestations of Maasai culture such as consumption of alcohol, bad language used at ceremonies and the worshipping of spirits. She approves, however, of the links which her husband has forged through his Maasai leadership activities and the social support provided through marriage arrangements and extended kin networks. It was evident from interviews that cultural norms were heavily debated in this household and that some accommodation between potentially conflicting positions was arrived at.

Varying claims on tradition and modernity are common in the construction of institutions through bricolage, a point well illustrated by Upperman's study of traditional irrigation in Tanzania where he shows how concepts of tradition and modernity are used in gendered contestations over access to resources in an irrigation system [Upperman, 2000]. But there are limits to negotiation, some norms being so deeply embedded that people would find it almost impossible to be discursively critical of them [Odgaard and Bentzon, 2001; Cleaver, 2000]. Indeed Douglas suggests that institutional arrangements are often so deeply embedded that it is almost impossible for people to subject them to conscious scrutiny: 'The high triumph of institutional thinking is to make the institutions completely invisible' [Douglas, 1987: 98]. Understanding institutional formation then requires us to try and uncover the deeply embedded principles which underlie people's choices and shape conscious decisionmaking, to identify the scope for flexibility in negotiating norms and practices.

VI. IMPROVISATION AND THE REINVENTION OF INSTITUTIONS

The concept of the crafting of institutions suggests that specific institutions are deliberately developed for particular functions. Thus an Irrigation Committee regulates the distribution of water and maintenance of furrows and a Grazing Association co-ordinates the use of grasslands. However, single purpose institutions are not favoured through processes of institutional bricolage. In the multiple processes of institutional evolution through bricolage, existing decision-making arrangements and relations of co-operation may be co-opted for new purposes. Such adapted, multi-purpose institutions abound in Usangu. For example, in study villages evangelical church choirs seem to be some of the most vibrant forms of associational life, with membership crossing ethnic, gender and livelihood divides. Members not only sing in church but may also function together as rotating credit groups, collective labour groups (also working as hired labour gangs on village works) and singers at 'traditional' social ceremonies and functions. Such embedded institutions combine productive and social functions, draw on both traditional and modern forms of interaction.

'Informal' institutions and decision-making about natural resource management are deeply culturally embedded. Livelihoods are not simply technically and economically rational sets of survival strategies but are clearly linked to ideas about a desirable way of life, to practices in relation to resources, to other people and to aspirations that are heavily loaded with symbolic meaning [Mehta *et al.*, 1999]. This has been elaborated in regard to water use [Mosse, 1997; Adams, 1997; Cleaver 2000]. However, cosmologies are not merely inherited as a static part of culture but are learnt and adapted in

the business of everyday life [Bourdieu, 1977]. Institutions formed as a result of bricolage in the multi-ethnic Usangu basin may be multi-cultural in origin, intersecting formal and informal, traditional and modern domains.

Mixing Traditional and Modern Arrangements

An example illustrates the mixed origins of institutions formed through bricolage. Farmers in Usangu commonly refer to 'traditional' smallholder rice irrigation systems (differentiating these from government run large schemes). However, this 'tradition' has a mixed and partly exotic provenance. Rice growing probably started in the mid nineteenth century, introduced by members of Arab trading caravans, whilst the irrigated gardens of the indigenous Sangu were observed by early explorers (personal communication: Rie Odgaard). 'Modern' rice irrigation was introduced into Usangu by Baluchi immigrants from Iran in the 1940s. 'Traditional' smallholder irrigation management draws on adapted committee structures introduced under government and NGO development projects (now mostly defunct) and on 'indigenous' collective labour arrangements. Whilst water co-operatives and associations were in some cases established in the past in order to formally claim water rights, many of these are non-functional, farmers perceiving them as unnecessarily bureaucratic [Gillingham, 1999]. However, roles, such as 'chairman' established under such projects, often remain. Conflicts over irrigation water are generally resolved between irrigators themselves, by reference to 'traditional' elders and (Sangu) customs, only if unresolvable are they referred to the Village Government and to Ward tribunals. Baluchis resolve disputes through reference to Islamic law or statutory rights [Maganga and Juma, 1999].

The evolutionary ebb and flow of irrigation management institutions, their mixed origins and the preference for socially embedded arrangements rather than bureaucratic ones all illustrate processes of institutional formation through bricolage.

Improvisation

The improvisatory, *ad hoc* and often intermittent nature of institutional arrangements formed through borrowing and adaptation are further illustrated by the example of the management of the water pump at Ukwaheri village. The example also illustrates how quickly formal institutional arrangements become adapted to social relationships and divisions.

Prior to the building of the 'SMUWC pump' in 1999 at Ukwaheri hamlet, severe water shortages were experienced here, sometimes resulting in the night-time stealing of water from other villages by Ukwaheri women. The water pump is managed under the village authorities and there is a six-member waterpoint committee and two specially trained mechanics. The committee has

three duties which were 'given' to them in their training: to maintain the pump in working order, to regulate water use and opening hours, to collect payments for use of the pump.

They have tried different forms of charging for water use. They tried charging a flat rate of 20 sh. per bucket, paid at the point of use, but this was problematic to enforce as it meant supervising water collection at all times. They also tried charging all adult users over 22 years old 1,000 sh. for the duration of the water-scarce season. However, people proved reluctant to pay these charges and expressed doubts over what happened to their money once collected. While Sukuma residents were willing payers, Sangu residents often refused to pay, expressing little trust in the (largely Sangu) committee and leadership. Nevertheless, by the middle of 2000, 170,000 sh. had been collected and deposited at the bank.

At the time of the research the committee were planning to introduce a different system of payment whereby houses are charged differentially according to their actual water use. Therefore a Sukuma household who may collect 30 buckets every two days will be charged more than a Sangu household who only collect five or six buckets a day. The committee claimed to have a good idea of approximate water use by household and therefore to be able to set such charges. They intended to charge for the dry season only – at other times of the year people collect from other sources and demand on the pump is reduced.

Potentially people can be fined for breaking water pump by-laws which relate to taking turns in water collection, keeping the surroundings clean and a prohibition on allowing animals to drink at the pump. No-one has, as yet, been fined for breach of these. However, the improvisatory nature of institution building, illustrated by this example, does not mean that any arrangement is possible or acceptable, new arrangements generally have to be awarded some social legitimacy, even if this involves inventing tradition.

Leakage of Meaning

Deeply embedded understandings of the 'right way of doing things' mean that people discursively draw on legitimising symbols (derived from the state, from culture and tradition, from the natural world), to cognitively anchor new institutional and social arrangements. In Douglas's words 'there needs to be an analogy by which the formal structure of a crucial set of social relations is found in the physical world, or in the supernatural world, or in eternity, anywhere so long as it is not seen as a socially contrived arrangement' [Douglas, 1987: 48].

Although claims on tradition can be seen as legitimising devices, tradition is not automatically accepted by all actors, nor is it necessarily sacrosanct. Paradoxically, in Usangu, the potential for questioning tradition on the one

hand and the general legitimacy of tradition on the other, means that cultural institutions may be 'borrowed' between ethnic groups, a key aspect of bricolage. The leakage of cultural rules and meanings across ethnic divides is well illustrated by the case of Mama N'Giriama, the caretaker of an important Sangu shrine, who conducts the rituals concerning the fertility and well-being of the *ihefu* and the people who live there. In her interpretation of the wishes of the ancestral spirit she emphasises incorporation and accommodation. She claims that there is a place for all on the *ihefu*, as long as people show proper respect and ask permission of the spirit to use it. The rituals she conducts appear to have become a multi-ethnic institution. For example, Sukuma and Masaai pastoralists (whose own ancestral spirits are based in distant lands of origin) may come and seek the blessing of the N'Giriama spirit so that their cattle do not get lost or stuck in the *ihefu*. Similarly people of different ethnic origins consult Mama N'Giriama for help in solving personal or health problems.

The Sungusungu – An Example of Bricolage

In processes of borrowing and adaptation the distinction between what is modern and what is traditional becomes blurred, tradition becomes reinvented. Additionally the line between formal organisation and socially and culturally embedded networks through which co-operation is forged become blurred. In Usangu we found an adaptation of traditional Sukuma militias of young men, to replace or supplement official Village Defence Committees. In several villages fear of cattle theft, the need to resolve potential competition over grazing and lack of confidence in government institutions led to the local establishment of a *Sungusungu* or cattle militia, borrowed from Sukuma customary defence organisations. Such institutions have become cross-ethnic, with ethnic Sangu and Masaai as well as Sukuma operating as 'commanders'.

The *Sungusungu* is made responsible by village consensus for cattle security and keeping order in the seasonal grazing lands. It operates on a basis of demarcation of roles between elders and youth, a practice common to all ethnic groups, with the youth acting as the foot soldiers and the elders acting as advisers on tactics, bestowers of charms and medicines, and dispensers of justice. This unofficial militia is considered by members to be formally accountable to (modern) village government, while the practices of its operation are largely based on socially embedded principles of reconciliation and conflict minimisation.

The *Sungusungu*, like many institutions formed through processes of bricolage, is multi-purpose. Villagers reported how they call on *Sungusungu* when facing problems requiring collective action, such as searching for a lost child, and use *Sungusungu* communication channels to disseminate messages around the village. One of the *Sungusungu* operating among seasonal grazing

camps at the *ihefu*, organises the disparate camps of young men (from widely dispersed villages) into units, the commander collects from them a seasonal subscription of cash and this is used as a common welfare fund to pay for a bus or bicycle to transfer a sick herder to his home area.

The benign nature of the *Sungusungu* that we find in Usangu is in contrast to more violent *Sungusungu* vigilante groups documented elsewhere in Tanzania. The diversity of insitutional forms and purposes all going under the name *Sungusungu* [Abrahams, 1998] further illustrates processes of bricolage and the adaptation of 'traditional' arrangements to new purposes.

The Transaction Costs of Bricolage

It is often claimed in institutional theory that socially and culturally embedded institutions economises on transaction costs as people can quickly draw on common norms in formulating collective arrangements [Mehta *et al.*, 1999]. However, whilst calling on familiar symbols and devices offers a sort of cognitive efficiency, it does not necessarily constitute a short cut to formulating legitimate institutions. In processes of institutional bricolage 'the categories of political discourse, the cognitive base of the social order are being constantly negotiated' [Douglas, 1987: 29]. Collective consensus, decision-making, forging and renegotiating norms, maintaining social networks, reputations and relations of trust are not easy processes, and require inputs of considerable time and energy. Community meetings are often lengthy, with an emphasis on achieving consensus rather than voting, and issues may only be resolved over weeks and months as some social consensus is gradually formed and the acquiescence of dissenters secured [Clever, 2000]. The time and effort involved in institutional bricolage are a particular burden to the poorest people who may be partially excluded as a result. Many interviewees from the poorest households in Usangu specified time constraints as their reason for non-participation in village decision-making.

An example of the effortful creation and maintenance of norms is seen in the priority given to avoiding or resolving conflicts, in constructing institutional mechanisms which emphasise reconciliation, forgiveness and an expectation of future co-operation.

VII. CONFLICT AVOIDANCE AND CO-OPERATION

The solidarity that makes collective action possible is something that has to be constantly worked at and neither socially embedded nor bureaucratic institutional structures obviate this. Here I use the example of the commonly held desire to avoid, minimise or peacefully resolve conflicts to illustrate how certain deeply embedded social values shape preferred institutional arrangements.

Project and policy approaches tend to see conflict as undesirable, as a breakdown in normal relations, something to be avoided or resolved as quickly as possible and optimistically insistent on perceiving co-operation as the norm and as 'the basis of village life' [Devitt, 1999; SMUWC, 2001a]. Perversely, though, bureaucratic institutional arrangements often emphasise the open confrontation of difference and the strict penalisation of non-conformers [Ostrom, 1990; SMUWC, 2001a, 2001b]. However, evidence of conflict and co-operation suggests a more complex picture than this. Conflict is both an integral part of normal life and something to be avoided or underplayed whenever possible.

A basic psychological dislike of conflict is illustrated by interviewees who identify the occasional conflicts with neighbours and kin as a major (if intermittent) source of stress in their lives, especially when associated with witchcraft. Additionally, where relations of reciprocity and institutions help to channel access to resources [Berry, 1989] then conflict must be avoided to ensure secure access to material livelihood assets. In Usangu, agricultural and pastoral families are networked through labour exchange, the use of draught power and by intermarriage as well as by church and club membership. But an over-emphasis on direct and instrumental reciprocity is misplaced. Relations of co-operation may be indirect and function across lifecourses and even generations as well as across localities. Additionally, many interviewees, when asked to identify the benefits of belonging to specific groups and associations, emphasised the opportunity for enjoyable social interaction, of social and psychological support, in addition to functional and productive benefits.

I suggest that values of social respect and conflict avoidance are deeply embedded and that these link moral behaviour to individual and community well-being (a form of moral-ecological rationality is explained further in Cleaver [2000]). The role of the supernatural (spirits, the ancestors and God) in securing the well-being of both humans and natural resources is strongly linked in cultural codes to people's behaviour, principles of respect and the desirability of peaceful coexistence.

Norms of conflict avoidance are common to all types of resource users in Usangu. Despite the rhetoric of high levels of conflict in Usangu and supposed irreconcilability of different cultures, Maganga [SMUWC, 2001d] found very few cases of conflicts over resources reaching court, reflecting a strong desire among people to resolve these at the lowest possible level. Values emphasising social respect and deference to elders are common to all social groups (if not always adhered to), but interviewees suggested that people do not merely inherit positions as leaders but also earn them through their ability to resolve conflicts and encourage harmonious relations with communities. Several interviewees had moved from their home area in order to avoid conflicts with neighbours and kin.

Celebrating Reconciliation, Reinforcing Co-operation

If collective action is to work, positive co-operation has to be reinforced, social capital built. The practice of actively embedding relations of co-operation in cultural and social life and of emphasising inclusive relations is common to the institutions formed through bricolage in Usangu. In direct contrast, literature on institutional design principles emphasises the need for impartial, rigorously enforced sanctions against non-co-operators [Ostrom, 1990, 1992]. Such principles, reproduced in policy documents [SMUWC, 2001b] emphasise confrontation and punishment rather than compromise and reconciliation and may erode rather than reinforce the social trust on which institutions depend.

The imperative towards emphasising co-operation even where conflict exists is strong. Evidence of potential conflict being minimised and turned towards co-operation instead is illustrated in accounts from Usangu and elsewhere in Tanzania [Mnzava, 2000; Maseruli, 2000]. These illustrate how communities (often hamlets and villages) may impose fines or penalties on those repeatedly offending against communal rules or failing to co-operate in communal work. Such penalties are only imposed when the social situation and extenuating circumstances of the offender are taken into account, a certain amount of 'social riding' being permitted. When extenuating circumstances are not taken into account, fines may be contested. In the relatively uncommon event of fines actually being levied, then the proceeds (money, livestock, household goods) are used to fund a celebration (a beer drink or feast) for those who did participate in the communal activity as well as the offender. According to informants, one of the purposes of this occasion is to 'celebrate forgiveness'.

It would be naive, of course, to present too rosy a picture of conflict avoidance and reconciliation. We have plenty of examples of mistrust between people (particularly over the management of funds), and norms of reconciliation may not benefit all. Maganga [SMUWC, 2001d] cites several cases of women filing for divorce through Ward Tribunals, possibly because they did not want to accept the reconciliatory mediation of local (male) elders at hamlet and village level. Moreover, certain people may be identified at village level as lazy, drunken or even witches and therefore unworthy of reconciliatory efforts. Despite undoubted limitations, however, I suggest that values prioritising conflict avoidance and reconciliation are deeply socially embedded and shape people's preferred arrangements for collective action and use of natural resources.

VIII. CONCLUSIONS

In this study I have outlined the concept of institutional bricolage in an attempt to strengthen our understanding of processes of resource management and

collective action. The introduction of new bureaucratic institutions or organisational arrangements are not necessarily robust and enduring, nor do they automatically ensure beneficial collective action and optimum resource use. Arrangements which rely on a blueprint derived from abstract and universalised 'design principles' may result in inadequate institutional solutions as they fail to recognise the depth of social and cultural embeddedness of decision-making and co-operative relations. Bureaucratic arrangements may be based on principles which bypass or contradict those inherent to local decision-making and co-operation, such as the minimisation of conflict. New bureaucratic institutions are unlikely to have evolved through a process of institutional bricolage and may be perceived by local people as costly, lacking in legitimacy and cumbersome. It is possible that such new institutions will gradually be subjected to a process of evolution, that over time processes of bricolage will ensure their redundancy or their adaptation to create more socially embedded arrangements.

In recognising the plurality of institutional arrangements and avoiding sterile dichotomies of traditional and modern, formal and informal, economic and social, we should beware of normatively attributing value to particular types of arrangement. There are indeed questions of how far bureaucratic institutions based on individual rights and the principles of modernity undermine social trust, relationships based on the ethics of care and mutual interdependence as suggested by Upperman's study [2000] of the formalisation of water rights in Tanzania. While bureaucratic arrangements are not necessarily inclusive, fair and emancipatory, socially embedded institutions may reproduce social divisions or gloss over inequality. Indeed it is possible that different kinds of institutions contribute to the same exclusionary processes.

However, plurality also creates opportunity, processes of bricolage, of borrowing, of institutional improvisation which may also create spaces for negotiation, contestation, and for different voices to be heard. Rather than seeing plural and *ad hoc* institutional arrangements as dysfunctional (as does much development policy [Berry, 1994]), we could see their very plasticity as providing scope for shaping social distribution and relationships in more equal and emancipatory directions.

If there is a solution to the problem of managing natural resources through institutions of collective action, I see it in finding ways in which newly introduced bureaucratic arrangements can complement or reinforce the positive aspects of socially embedded arrangements. Where natural resource management requires strengthening, we need interventions based on an understanding of the content, underlying principles and social effects of institutions, not merely their visible form. I see great potential for effective development interventions which recognise the importance of processes of

bricolage, rather than simply emphasising their manifestation as structures and outcomes, deliberately crafted. The knowledge gained through an archaeology of institutions should not delude us into thinking that institutions are wholly amenable to design. We cannot predict exactly how newly introduced arrangements will become revised, adapted and socially embedded over time, or abandoned and forgotten, through processes of institutional bricolage.

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